

Availability Sets & Virtual Machines

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Date: 26/05/2026

Objective

1. Create an Availability Set (AS)
2. Create a Virtual Machine (VM) in that Availability Set
3. Add User Data script in the VM during creation
4. Verify that the script ran properly
5. Reset the password of the VM
6. Change the instance size of the VM

Task 1: Create an Availability Set (AS)

Azure Portal → Search 'Availability Sets' → Create new AS with required configuration (MyAS).

The screenshot displays the Azure portal interface for an Availability Set named 'MyAS'. The top navigation bar includes the Microsoft Azure logo, a search bar, and user profile information. The left sidebar shows the 'Overview' tab selected, with links to Activity log, Access control (IAM), Tags, Resource visualizer, Settings, Automation, and Help. The main content area is titled 'MyAS Availability set' and includes buttons for 'Create a VM replica', 'Suggest VM sizes for hosting my application', and 'Recommend an Azure VM size for my workload'. Below this, the 'Essentials' section provides key details: Resource group (test), Location (Central India), Subscription (Azure for Students), and Subscription ID (772ad338-a391-45c6-a1e7-591785004040). To the right, configuration parameters are listed: Fault domains (2), Update domains (5), Virtual machines (0), Managed (Yes), and Colocation status (N/A). A 'JSON View' link is available. At the bottom, a table for 'Search virtual machines' is shown with columns for Name, Status, Colocation status, Fault Domain, and Update Domain, but it currently displays 'No results'.

Task 2: Create a VM in the Availability Set

Created a new Virtual Machine (MyAzureVM) and selected the above Availability Set (MyAS) during the Basics configuration tab.

The screenshot shows the 'Create a virtual machine' page in the Microsoft Azure portal. The 'Basics' configuration tab is active. At the top, there are navigation links: 'Home > Compute infrastructure | Virtual machines', a search bar, and a Copilot button. Below the navigation, there are three help links: 'Help me create a low cost VM', 'Help me choose the right VM size for my workload', and 'Help me create a VM optimized for high availability'. A warning message states: 'Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.' The 'Project details' section includes 'Subscription' (Azure for Students) and 'Resource group' (test). The 'Instance details' section includes 'Virtual machine name' (MyAzureVM), 'Region' ((Asia Pacific) Central India), and 'Availability options' (Availability set). A blue information box suggests creating a virtual machine scale set. At the bottom, there are navigation buttons: '< Previous', 'Next : Disks >', and 'Review + create'.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me create a low cost VM Help me choose the right VM size for my workload Help me create a VM optimized for high availability

Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource group * test

Create new

Instance details

Virtual machine name * MyAzureVM

Region * (Asia Pacific) Central India

Deploy to an Azure Extended Zone

Availability options Availability set

Based on your input, you might want to consider creating this resource as a virtual machine scale set, which allows you to manage, configure and scale load balanced virtual machines. [Create as VMSS](#)

< Previous Next : Disks > Review + create

Give feedback

Task 3: Add User Data in VM During Creation

Under the Advanced tab of VM creation, added a startup script in the 'Custom data' / 'User Data' field to install and start Apache2.

The screenshot shows the 'Create a virtual machine' page in the Microsoft Azure portal, specifically the 'Advanced' configuration tab. The 'Custom data and cloud init' section is visible. It includes a description of custom data and a text area for 'Custom data' containing a bash script to install and start Apache2. Below the text area, there is a note about cloud-init processing custom data. The 'User data' section is also visible, with a checkbox to 'Enable user data' which is checked. At the bottom, there are navigation buttons: '< Previous', 'Next : Tags >', and 'Review + create'.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me create a low cost VM Help me choose the right VM size for my workload Help me create a VM optimized for high availability

VM applications contain application files that are securely and reliably downloaded on your VM after deployment. In addition to the application files, an install and uninstall script are included in the application. You can easily add or remove applications on your VM after create. [Learn more](#)

Select a VM application to install

Custom data and cloud init

Pass a cloud-init script, configuration file, or other data into the virtual machine **while it is being provisioned**. The data will be saved on the VM in a known location. [Learn more about custom data for VMs](#)

Custom data

```
#!/bin/bash
apt update -y
apt install apache2 -y
systemctl start apache2
```

Custom data on the selected image will be processed by cloud-init. [Learn more about custom data for VMs](#)

User data

Pass a script, configuration file, or other data that will be accessible to your applications throughout the lifetime of the virtual machine. Don't use user data for storing your secrets or passwords. [Learn more about user data for VMs](#)

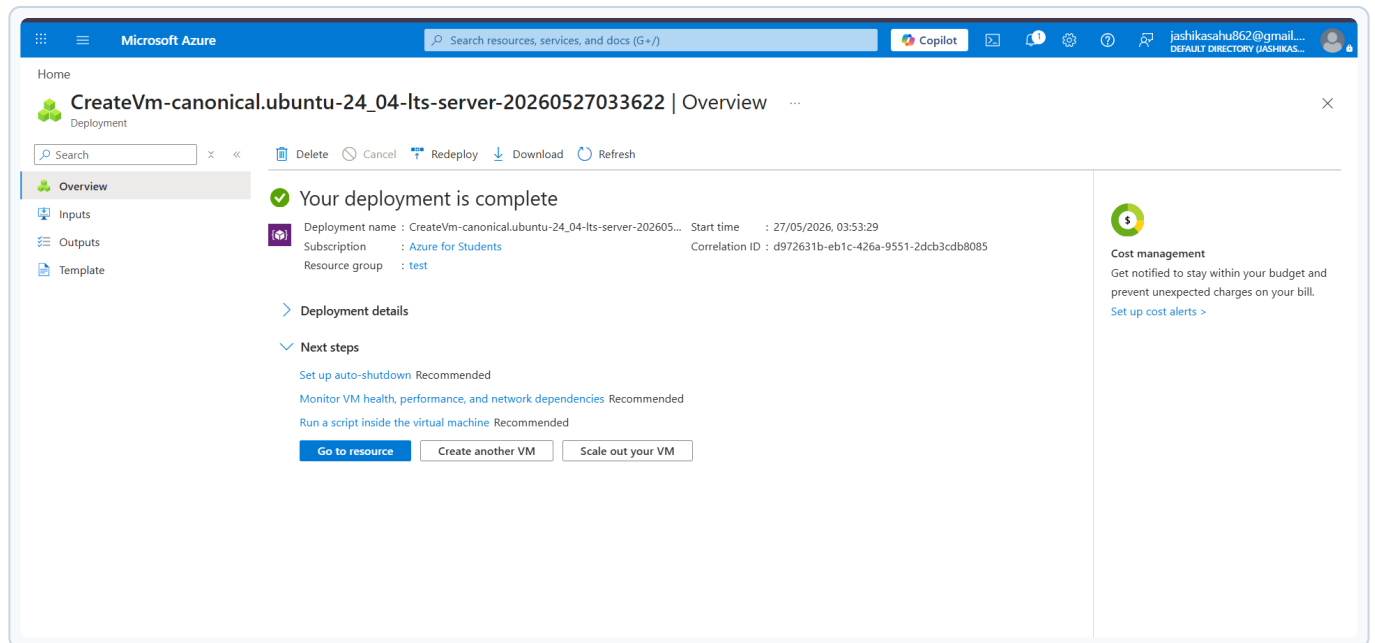
Enable user data ☒

< Previous Next : Tags > Review + create

Give feedback

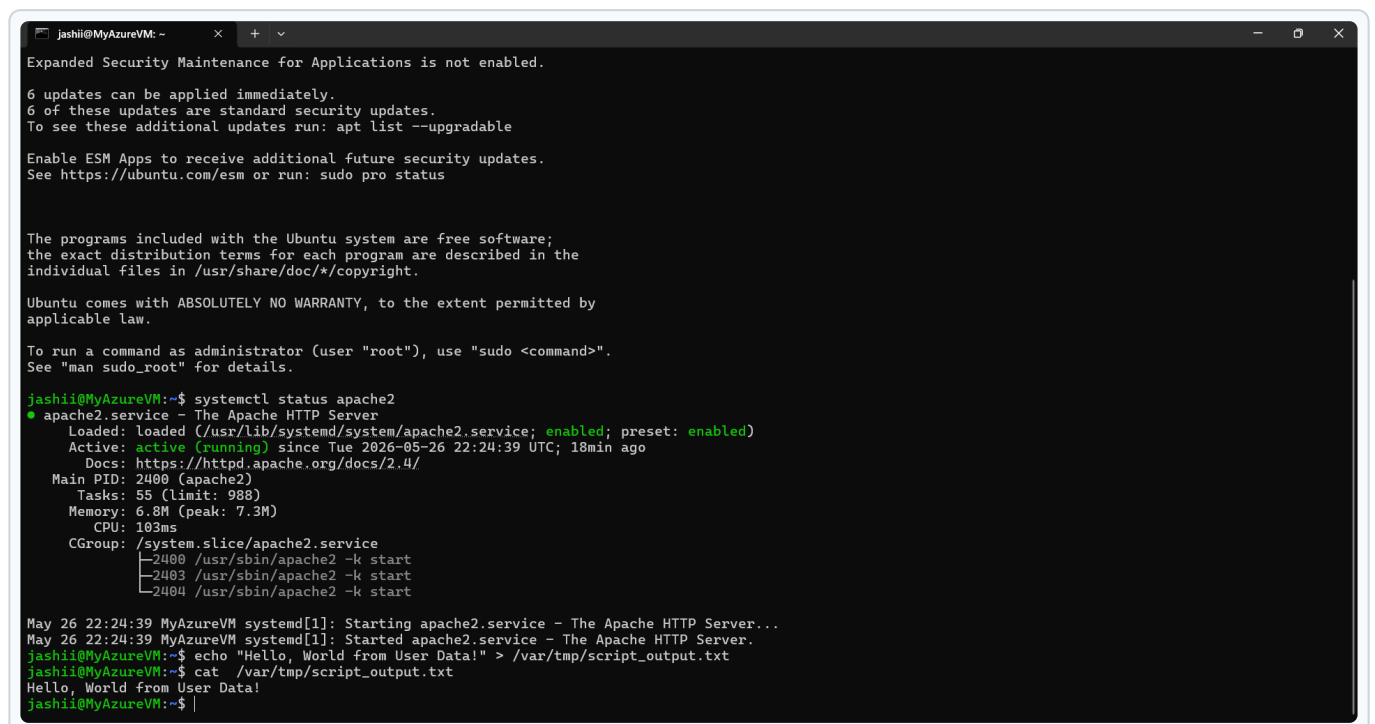
VM Deployment Verification

Monitored the process until the template deployment successfully completed in the resource group.



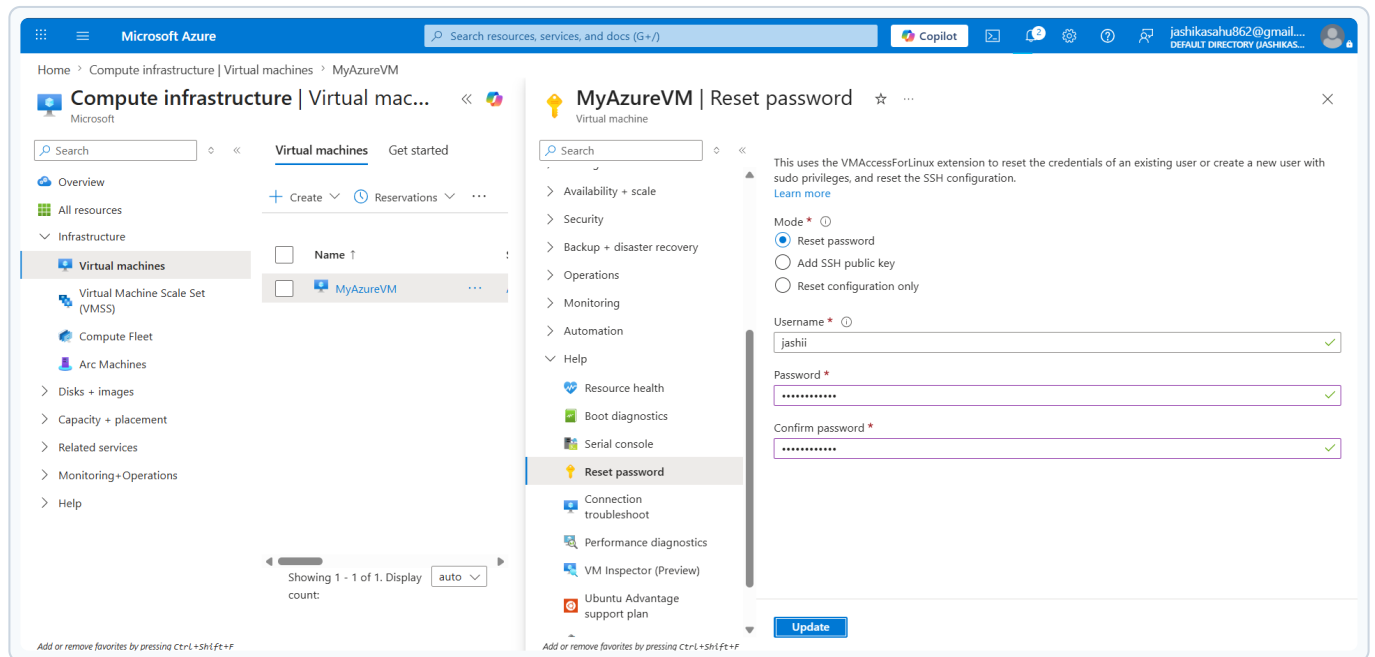
Task 4: Verify Script Ran Properly

After VM deployment, connected via SSH and verified that the user data script executed successfully by checking Apache2 service status and verifying script output.



Task 5: Reset the Password of VM

Navigated to VM → Help → Reset Password. Used the 'Reset password' option to update credentials for the user 'jashii'.



Task 6: Change the Instance Size of VM

Navigated to VM → Settings → Size. Selected a new VM size (e.g., Standard_B2as_v2) and applied the resize configuration.

